

### 0.0.1 Boundary pressure error convergence

The volume pressure norm is the primary convergence measure for the Stokes benchmarks, but pressure errors on the inner and outer boundaries provide a more local test of the pressure trace. This is relevant for geodynamics applications because boundary tractions, radial normal stresses, and free-slip or zero-slip boundary terms all depend on how accurately the discrete solution represents pressure near curved surfaces. For a boundary  $\Gamma \in \{\Gamma_{\text{inner}}, \Gamma_{\text{outer}}\}$ , we denote the absolute pressure-trace error by

$$E_{L_2, \Gamma}(p) = \left( \int_{\Gamma} |p_h - p^*|^2 d\Gamma \right)^{1/2}, \quad (1)$$

and, where the analytical boundary pressure has a nonzero  $L_2$  norm, the relative boundary pressure error by

$$E_{L_2, \Gamma}^{\text{rel}}(p) = \left( \frac{\int_{\Gamma} |p_h - p^*|^2 d\Gamma}{\int_{\Gamma} |p^*|^2 d\Gamma} \right)^{1/2}. \quad (2)$$

As in the volume pressure comparison, the pressure gauge is fixed before evaluating errors so that the numerical and analytical pressures are compared in the same mean-pressure convention.

The Thieulot–Puckett annulus benchmark gives a smooth manufactured solution on a curved annulus. For the  $P_2 \times P_1$  Taylor–Hood discretisation, the inner and outer boundary pressure errors decrease close to the expected  $O(h^2)$  trend, although the two boundaries have different error constants. This difference is not unexpected: the inner and outer curves have different radii, different geometric quadrature paths, and different sensitivity to the linear approximation of the curved annulus boundary.

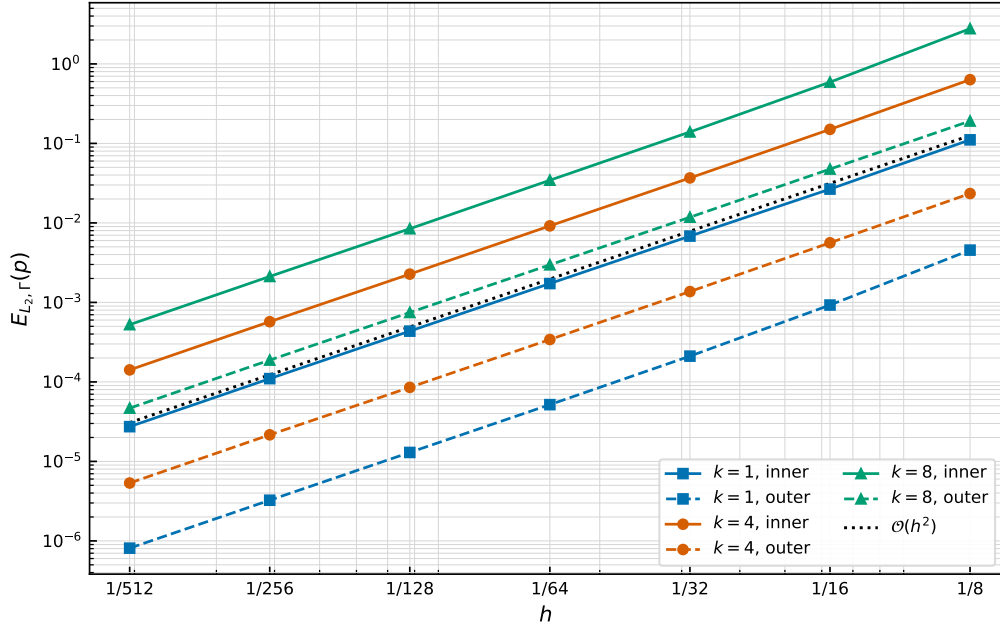


Figure 1: Absolute boundary pressure error convergence for the Thieulot–Puckett annulus benchmark using the  $P_2 \times P_1$  discretisation. Solid curves denote the inner boundary and dashed curves denote the outer boundary.

The Kramer annulus benchmark separates smooth volumetric forcing from delta-function forcing on an internal circular interface. The boundary pressure errors converge close to second order

for the smooth cases. The delta-function cases also show stronger boundary pressure convergence than the corresponding volume pressure norm because the reduced regularity is localized at the internal interface rather than on the annulus boundaries.

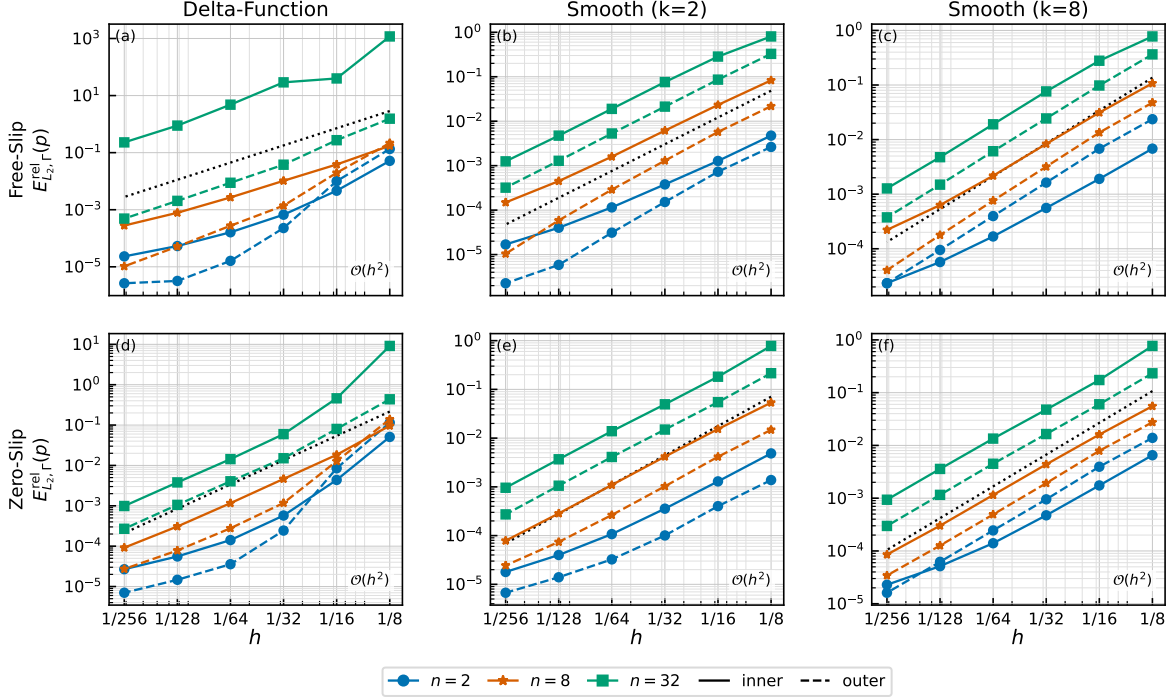


Figure 2: Relative boundary pressure error convergence for the Kramer annulus benchmark. Solid curves denote the inner boundary and dashed curves denote the outer boundary.

The Thieulot spherical-shell benchmark gives the corresponding smooth three-dimensional test. The boundary-pressure panel shows that the absolute pressure trace errors for both  $m = -1$  and  $m = 3$  decrease systematically with refinement. These errors are shown together with radial normal-stress errors because the stress diagnostic combines pressure and velocity-gradient errors at the same curved spherical boundaries.

For the Kramer spherical-shell benchmark, the current combined boundary figure reports boundary velocity and radial normal-stress convergence for the free-slip delta-function case. A boundary pressure-trace figure is not part of the current spherical Kramer article outputs. The radial normal-stress diagnostic is nevertheless included here because it is the closest available boundary pressure-sensitive quantity:  $\sigma_{rr}$  contains the pressure contribution and therefore tests pressure recovery together with velocity-gradient recovery on the spherical boundaries.

Overall, the boundary diagnostics complement the volume  $L_2$ -norm pressure errors. They isolate the pressure trace on the surfaces where boundary conditions and stress diagnostics are evaluated. Smooth benchmark cases recover approximately second-order pressure-trace convergence for the  $P_2 \times P_1$  pair, while singular forcing primarily affects the volume pressure norm through reduced regularity at the internal interface. The persistent difference between inner- and outer-boundary error constants should therefore be interpreted as a geometric and boundary-evaluation effect rather than as a failure of pressure normalisation.

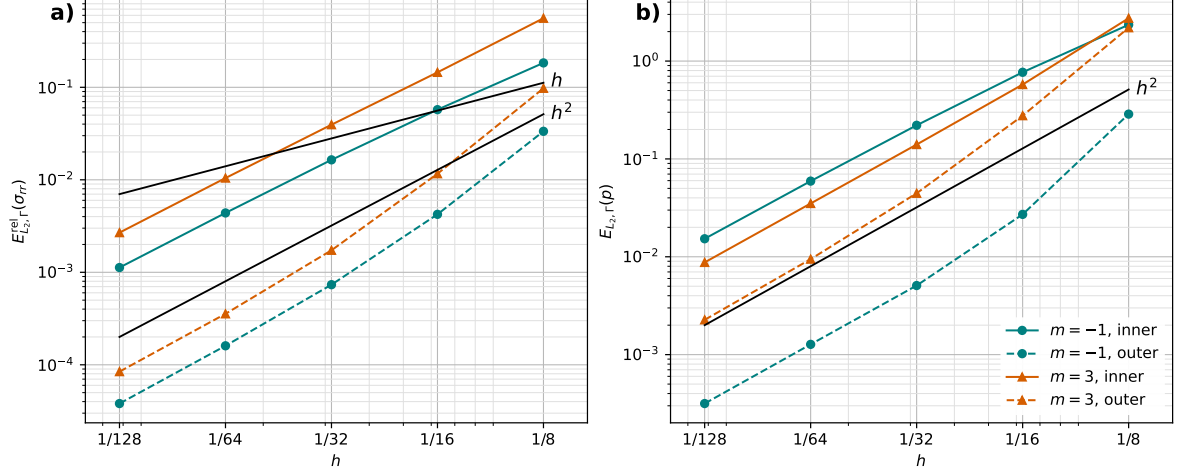


Figure 3: Boundary radial normal-stress and absolute boundary pressure error convergence for the Thieulot spherical-shell benchmark. The right panel shows  $E_{L2,\Gamma}(p)$  on the inner and outer spherical boundaries.

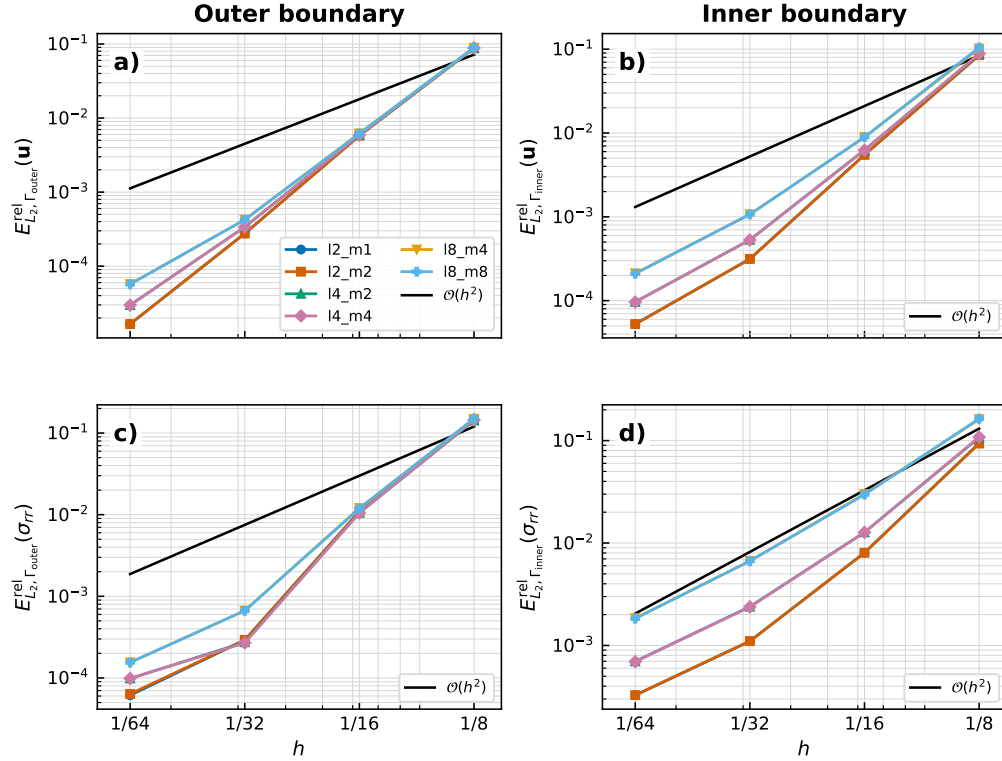


Figure 4: Boundary velocity and radial normal-stress convergence for the Kramer spherical-shell benchmark. This figure is included as the available pressure-sensitive boundary diagnostic for the spherical Kramer case; the current article outputs do not include a separate boundary pressure-trace convergence plot.